

Theorem 7.46: 3SAT reduces to Hamiltonian Cycle in a directed graph (so HC is NP-Hard).

Proof: We begin with a 3SAT formula

$$\Psi = (a_1 \vee b_1 \vee c_1) \wedge (a_2 \vee b_2 \vee c_2) \wedge \dots \wedge (a_k \vee b_k \vee c_k).$$

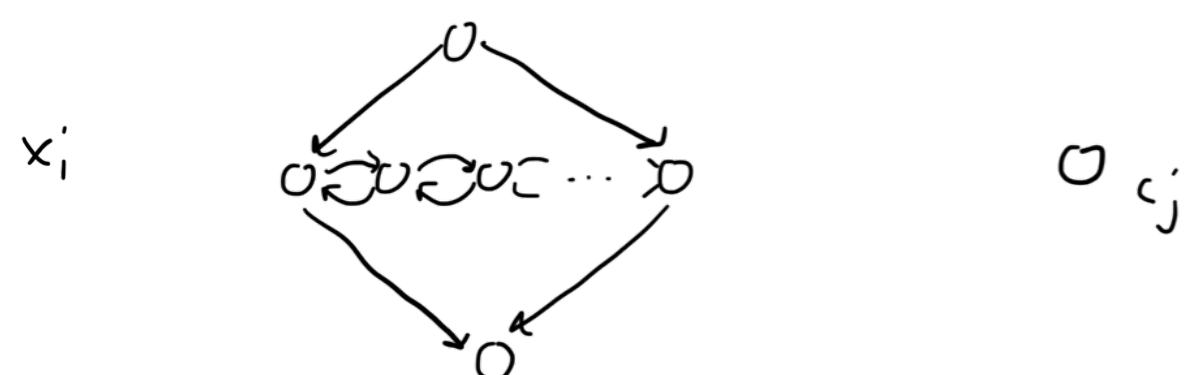
We let each a, b, c denote a literal x_i or $\bar{x}_i$.

Let $x_1, \dots, x_k$ be k variables of Ψ .

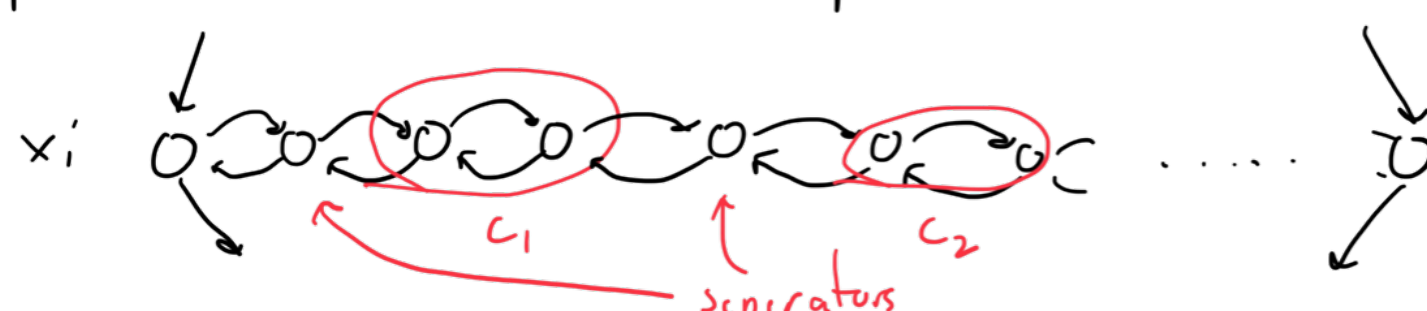
Ex: $\Psi = (x_1 \vee x_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee x_2 \vee x_3).$

We will convert Ψ into a graph G that has a HC iff Ψ is satisfiable.

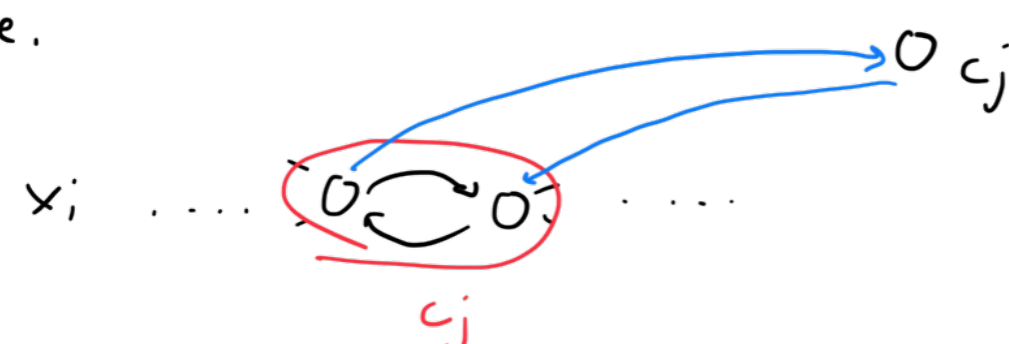
We convert each variable x_i into a "diamond-like" gadget. Each clause of Ψ is converted to a single vertex c_j ($1 \leq j \leq k$).



Each diamond has a horizontal row of $3k+3$ vertices. Each c_j is represented via a pair, with an extra separator vertex.



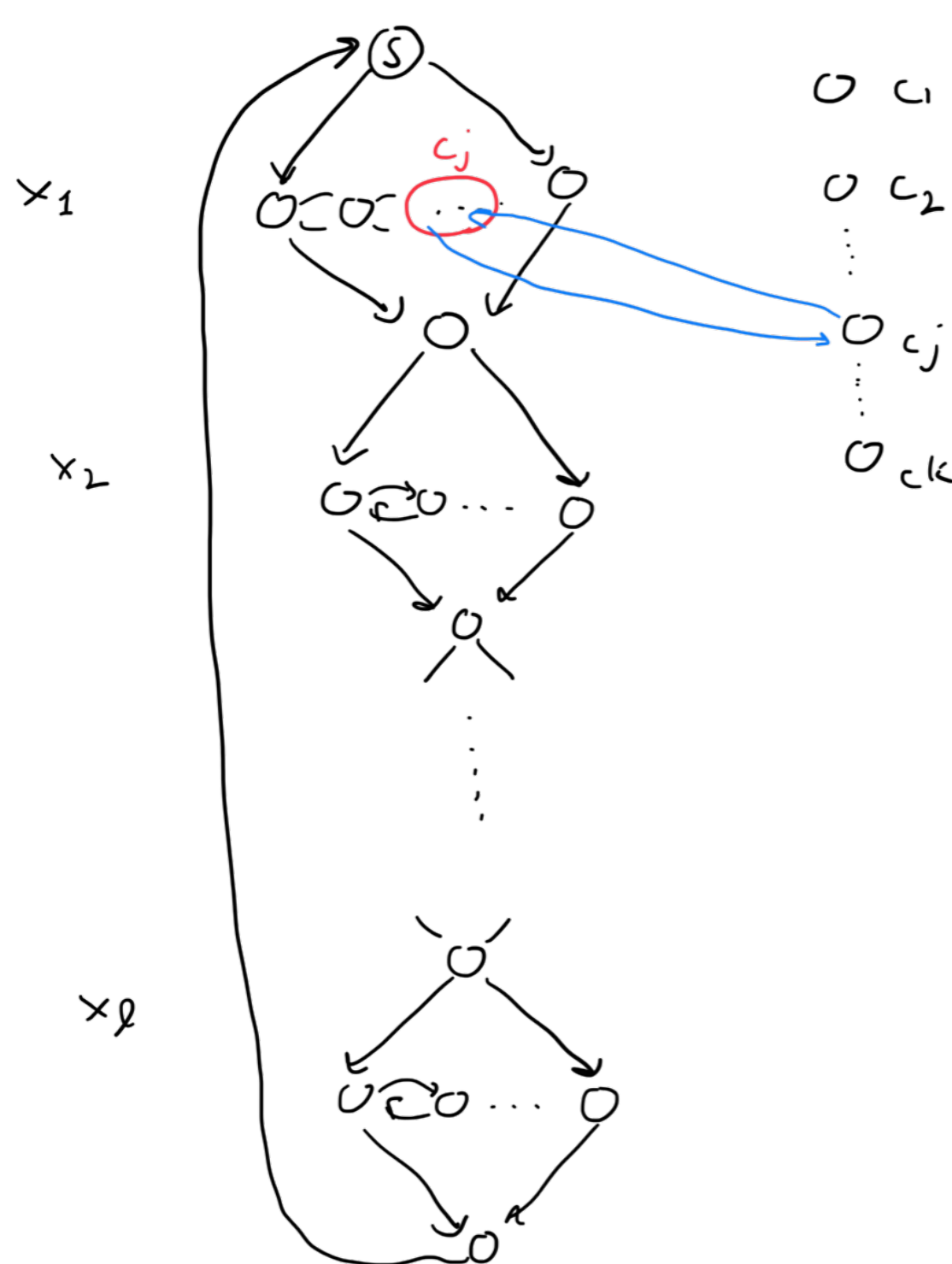
If x_i appears in clause c_j of Ψ , then 2 edges are added to G from the j -th pair of the i -th diamond to the j -th clause.



If $\bar{x}_i$ appears in clause c_j , we add 2 edges in the reverse direction.

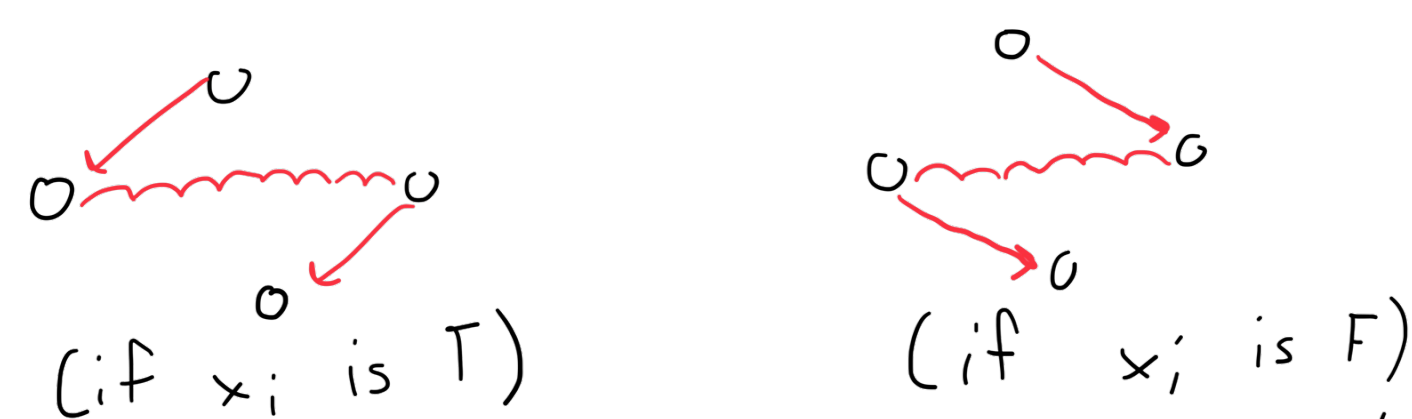


We end up with G as such:



(sufficiency)

$\Rightarrow$ Suppose Ψ is satisfiable. We can demonstrate a HC in G . Beginning from s , we go through each diamond and end at t . There are 2 ways to go through a diamond.



We can easily update the path to include the clause vertices via a detour. That is, if $x_i = T$, we take the 2 edges to include c_j via the c_j pair in the i -th diamond.

(necessity)

$\Leftarrow$ Suppose G has a HC P . We will show it satisfies Ψ . If P goes left-to-right in diamond i , then we assign True to the corresponding $x_i \in \Psi$. If P goes right-to-left in diamond i , then we assign $x_i = \text{False}$ in Ψ . Since each clause appears in P , we can determine which literals in the corresponding clause are True. $\square$

Corollary: HC in a directed graph is NPC.

Proof: Let G be a directed graph. We are given a cycle P . We can easily check whether P is a HC in $O(n)$ by iterating on all vertices in P and comparing $|P|$ to $|V|$.

This shows that HC is in NP.

We previously proved it is NP-Hard.

Thus, it is NPC. $\square$

